Calculus(1)

WORKSHEET #1

ID : _____

Name : _____

※ Show the details of your work.

Ch2.

Q1. Find an equation of the tangent line to the curve at the given point.

(a) $y = x^3 - 3x + 1,\ (2,3)$

(b) $y = \sqrt{x^3 + 3x}\ ,\ (1,2)$

(c) $y = \dfrac{2x+1}{x+2},\ (1,1)$

Q2. If the tangent line to $y = f(x)$ at $(9,3)$ passes through the point $(0,2)$, find $f'(9)$ and $f(9)$.

Q3. Find $f'(a)$, if

(a) $f(x) = x^{-2}$

(b) $f(x) = \dfrac{4}{\sqrt{1-x^2}}$

(c) $f(x)=\dfrac{x-1}{x+1}$

Q4. At what points do the graphs of the functions have horizontal tangent lines?

(a) $f(x)=x^2+4x-1$

(b) $g(x)=x^3-3x$

Q5. Find the derivatives of the function

$$y=\frac{(x+1)(x+2)}{(x-1)(x-2)}$$

Q6. Determine whether $f'(0)$ exists if

$$f(x)=\begin{cases} x\sin\dfrac{1}{x} & \text{if } x\neq 0 \\ 0 & \text{if } x=0 \end{cases}$$

Q7. Find an equation of the tangent line to the curve at the given point.

(a) $y=\dfrac{x^2-1}{x^2+x+1},\ (1,0)$

(b) $y=\cos x-\sin x,\ (\pi,-1)$

Q8. Find $f''(\pi/4)$ if $f(x)=\sec x$.

Q9. If $f(3)=4, g(3)=2, f'(3)=-6, \quad g'(3)=5,$ find

(a) $(f+g)'(3)$

(b) $(fg)'(3)$

(c) $(f/g)'(3)$

Q10. Find all points on the graph of the function $f(x)=\sin^2 x+2\sin x$ at which the tangent line is horizontal.

Q11. If $g(x)=f(xf(xf(x)))$, where $f(1)=2$, $f(2)=3$, $f'(1)=4$, $f'(2)=5$, $f'(3)=6$, $f'(4)=7$, find $g'(1)$.

Q12. If $f(x)+x^2[f(x)]^3=10$ and $f(1)=2$, find $f'(1)$.

Q13. Use implicit differentiation to find an equation of the tangent line to the curve $x^2+xy+y^2=3$ at $(1,1)$.

Q14. The length of a rectangle is increasing at a rate of $8\mathrm{cm/s}$ and its width is increasing at a rate of $3\mathrm{cm/s}$. When the length is $20\mathrm{cm}$ and the width is $10\mathrm{cm}$ how fast is the area of the rectangle increasing?

Q15. Suppose $4x^2+9y^2+36$, where x and y are functions of t. If $dy/dt=1/3$, find dx/dt when $x=2$ and $y=2\sqrt{5}/3$.